**SIMATS ENGINEERING**
**ASSESSMENT TOOL 2: Case Study**

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| **COURSE NAME:** | Cloud Computing and Big Data Analytics | **COURSE CODE:** | CSA15 |

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**COURSE OUTCOME COVERED**
**CO3:** *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 3

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**Code of Conduct**
I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____________________

**Assessment Tasks**
**Case Study**
1. **Case Study 1: Smart Retail Inventory Management System**

A retail chain implements a cloud-based inventory system connected to store devices.Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.
2. **Case Study 2: Cloud-Based Document Collaboration System**

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.
Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.
3. **Case Study 3: Online Food Ordering Platform**

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients.
Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.
Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability.

Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

**Case Study Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Understanding of Case | 25% | Clearly identifies all technical and business requirements | Identifies most requirements | Basic understanding only | Poor understanding of case |
| Application of Concepts | 25% | Applies suitable cloud concepts and tools effectively | Good application with minor issues | Partial application | Weak or incorrect application |
| Analysis | 20% | Strong analysis with clear trade-off reasoning | Reasonable analysis | Limited analysis | Minimal analysis |
| Solution Design | 20% | Solution is feasible, practical, and well justified | Reasonable solution with minor gaps | Basic solution | Unclear solution |
| Clarity | 10% | Well-structured and easy to follow | Mostly clear | Somewhat unclear | Poorly organized |

# Questions:

## Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

1. **Cloud Service Models Used:**

    - **IaaS (Infrastructure as a Service):** Provides virtual machines, networking, firewalls, and computing infrastructure.

- **PaaS (Platform as a Service):** Used for analytics, demand forecasting, and application development.
- **SaaS (Software as a Service):** POS dashboards and inventory applications can be accessed by users through the cloud.

2. **Clients:**

   - POS (Point of Sale) systems
   - Store devices
   - Management dashboards

3. **Communication:**

   Clients communicate with centralized cloud services using **Web APIs**.

4. **Cloud Storage:**

   Stores:

   - Product information
   - Transaction records
   - Analytics datasets

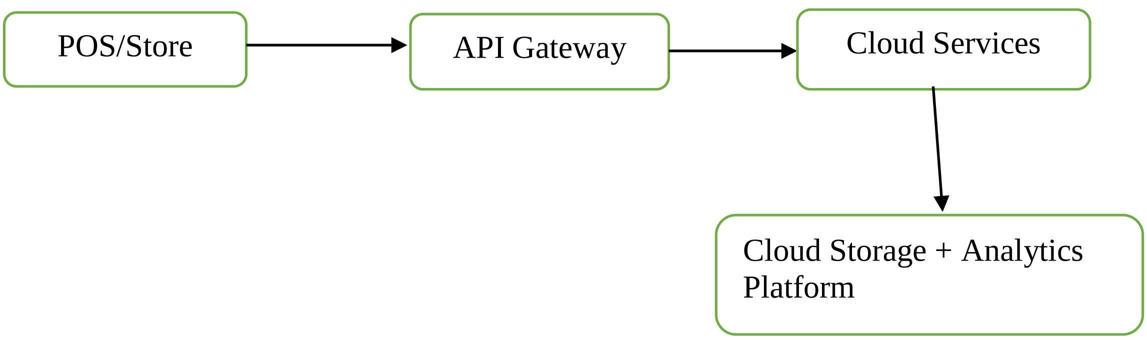
5. **Security:**

   - VPN
   - Firewalls
   - API Gateway
   - Authentication and authorization

6. **Benefits:**

   - Real-time inventory tracking
   - Demand forecasting
   - Easy scalability
   - Centralized data management
   - Improved security and availability

7. **Overall Architecture:**



## Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

### 1. Cloud Service Model

The system mainly follows **SaaS (Software as a Service)** because users access the document collaboration software through web browsers without managing the underlying infrastructure.

### 2. Clients

- Web browsers
- Computers
- Mobile devices
- Multiple users

### 3. Cloud Storage

Documents are stored in **distributed cloud storage**.

It provides:

- Data availability
- Scalability
- Backup
- Version control

### 4. Communication

**Web APIs** synchronize documents between different users and devices.

### 5. Version Control

Version control maintains different versions of documents and allows users to track or restore previous versions.

### 6. Security

- Data encryption
- Identity management
- Authentication
- Access permissions
- Authorization

### 7. Networking

Global cloud infrastructure provides:

- Low latency
- High availability
- Reliable communication

### 8. Overall Architecture


## Case Study 3: Online Food Ordering Platform

### 1. Cloud Service Models Used

- **IaaS:** Virtual machines provide computing infrastructure.
- **PaaS:** Managed Kubernetes provides a platform for deploying and scaling applications.
- **SaaS:** The food-ordering application is provided to customers through web/mobile clients.

### 2. Clients

- Web browsers
- Mobile applications

### 3. Communication

Frontend clients communicate with backend services using **RESTful Web APIs**.

### 4. Cloud Storage

Stores:

- Menu images
- Invoices
- Customer information
- Order details

### 5. Infrastructure

- Virtual Machines

- Managed Kubernetes
- Cloud networking

Kubernetes helps with:

- Scalability
- Load distribution
- Fault tolerance
- High availability

**6. Security**

- Authentication tokens
- HTTPS
- Role-Based Access Control (RBAC)

**7. Benefits**

- Supports many users
- Easy scaling during high demand
- Secure transactions
- Reliable service
- Fast access through web and mobile applications

**8. Overall Architecture**